

# Digital Dividend and UPI Adoption among Informal Vendors

## Evidence from the B.T. Road Corridor

**Bachelor of Statistics (Honours)**  
*class of 2023–26*

Indian Statistical Institute, Kolkata

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# Presentation Roadmap

- Introduction and Background
- Survey Design and Data
- Descriptive and Exploratory Analysis
- Inferential Analysis
- Conclusions

# Introduction and Background

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# Background and Motivation

## The Indian Digital Payments Revolution

- UPI launched by NPCI in **April 2016**
- Direct bank-to-bank transfers via mobile apps and QR codes
- Accelerated after **demonetisation (Nov 2016)**
- QR-code boards now common even among informal vendors

# Background and Motivation

## Why Micro-Level Evidence?

- Macro data (transaction volumes) do not capture *everyday economic realities*
- Street vendors face uncertainty: daily flow, weather, seasonality
- Need to examine transaction friction, customer retention, revenue stability

# Background and Motivation

## Research Gap

Existing studies focus on:

- Macroeconomic trends
- Banking infrastructure
- Consumer behaviour

**Missing:** Quantitative evidence on how UPI affects *roadside vendors*

## Study Area: The B.T. Road Corridor

- **Barrackpore Trunk Road** – one of the most commercially active transport routes in northern Kolkata
- Stretch: **Sinhi to Dunlop**
- Key localities: Bonhooghly, GLT Road, Dunlop Bridge, Sinhi More
- Continuous movement of commuters, students, industrial labourers

### Vendor Landscape

- Large concentration of *informal-sector* vendors
- Products: tea, snacks, fruits, garments, footwear, mobile accessories
- High-frequency transactions  $\Rightarrow$  efficiency is critical

## Study Area: Preliminary Recce Findings

### Preliminary Recce Findings

- Digital payment acceptance is **nearly universal**
- QR-code systems observed across all scales
- **Considerable variation** in *intensity* of UPI usage

#### Sample Size

**129** respondents

Systematic random sampling



# Project Objectives

## 1. Customer Volume and UPI Intensity

Test whether vendors with high UPI intensity serve significantly more customers than low-intensity vendors.

## 2. Sales Loss Due to Change Shortage

Estimate revenue losses from cash change-shortage and investigate whether UPI mitigates these transaction-related losses.

## 3. Revenue Volatility and Digital Transactions

Estimate short-term daily revenue volatility and determine if higher UPI intensity lowers volatility, controlling for goods type and location.

## Project Objectives (Contd.)

### 4. **Vendor Perception and Behavioural Response**

Examine perceptions on transaction speed, financial tracking, savings behaviour, and digital trust across demographics.

### 5. **Drivers of UPI Adoption**

Identify principal motivating factors (customer demand, competition, convenience) and structural fee arrangements.

## Ethical Considerations and Survey Protocol

- Survey involved direct interaction with informal-sector street vendors.
- Questions covered:
  - Earnings and income patterns
  - Transaction practices
  - Financial behaviour
- Special emphasis placed on:
  - Informed participation
  - Comfort of the respondent
  - Privacy protection
- Survey designed to ensure respectful and trustworthy data collection.

# Ethical Standards Followed (Contd.)

## 1. Voluntary Participation

- Respondents could refuse questions or withdraw anytime.
- No penalties or consequences for non-participation.

## 2. Sensitive Handling of Income Questions

- Income data collected only for academic purposes.
- Approximate/range-based responses encouraged.
- Questions asked respectfully to minimise discomfort.

## Ethical Standards Followed (Contd.)

### 3. **Protection of Identity and Privacy**

- No Aadhaar, bank details, or phone numbers collected.
- Names recorded only with explicit consent.

### 4. **Academic Identification of Surveyors**

- Surveyors introduced themselves as students from ISI.
- Institutional identity cards carried during fieldwork.

## Ethical Standards Followed (Contd.)

### 5. Political Neutrality

- Avoided political or ideological questions.
- No questions on unions or labour disputes.

### 6. Confidential Data Handling

- Information used strictly for academic research.
- No public sharing outside study scope.

# Survey Design and Data

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# Design Philosophy

## Three Guiding Questions

1. **Who is the vendor?**
2. **How is UPI used?**
3. **What difference does UPI make?**

### Categorical Background

- Gender, age group
- Type of goods sold
- Location, mode of operation

### UPI Adoption & Outcomes

- UPI acceptance, share of users
- Fee arrangements
- Customer flow, earnings, revenue variation

### Perceptual Variables

- Transaction speed
- Ease of tracking
- Digital trust, preference

*Final section: Surveyor internal consistency check to flag logical contradictions*



## Questionnaire Structure

| Sec. | Theme                | Key Variables                                                                                                               |
|------|----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1    | Vendor Profile       | Gender, age, goods type, location, years in business, mode of operation                                                     |
| 2    | UPI Usage & Adoption | Adoption date, reasons, proportion of customers making digital payments, UPI-related fees, occurrence of technical problems |
| 3    | Sales & Revenue      | Ticket size, daily customers, revenue stability, best/worst day earnings, last-hour customers                               |
| 4    | Perceptions          | Transaction speed vs. cash, customers lost (change/UPI), digital record worry, mode preference                              |
| 5    | Surveyor Check       | UPI/cash share consistency, QR visibility, interview atmosphere                                                             |

# Sampling Strategy

## Reconnaissance Survey

- Two teams walked assigned segments
- Mapping sheet: location, language, category, visible UPI status
- Ensured coverage and baseline estimate

## Pilot Survey

- **15** vendors in Dunlop Market
- Response: satisfactory
- Key observation: owners uncomfortable disclosing daily earnings

## Final Survey

- **Systematic random sampling**
- Two teams per road (each side)
- First shop random among first 5; then every 5th shop
- Average duration: **10 minutes**

# Data Pre-Processing

## Cleaning Steps

1. Standardized location labels (e.g., “KFC” vs “near KFC”)
2. Flagged implausible values (e.g., “13 hours” business age)
3. Checked last-10-customers column: totals far from 10 flagged
4. Income ranges converted to midpoints:
  - “x-y”  $\rightarrow$  midpoint
  - “above x”  $\rightarrow > x$
  - “below x”  $\rightarrow 0.75x$  or  $x$  (depending on context)

## Data Pre-Processing: Missing Values and Encoding

### Missing Value Treatment

- Isolated entries: imputed with *column mean* (e.g., fruit shops)
- Multiple missing values in a row: retained as NA

### Categorical Encoding

- Yes/No  $\rightarrow$  1/0
- Complete coding in Appendix A

#### Note

Business duration approximated to nearest year.

# Descriptive and Exploratory Analysis

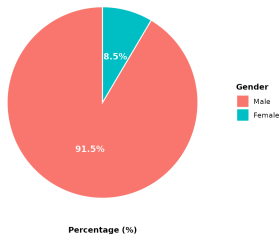
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# Demographic Profile: Gender and Age

## Gender Distribution

Gender Distribution of Street Vendors

n = 129 respondents

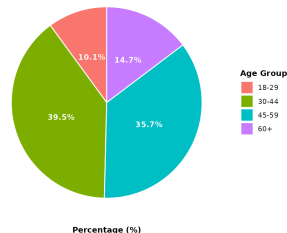


**Male** 91.5%  
**Female** 8.5%

## Age Distribution

Age Group Distribution of Street Vendors

n = 129 respondents



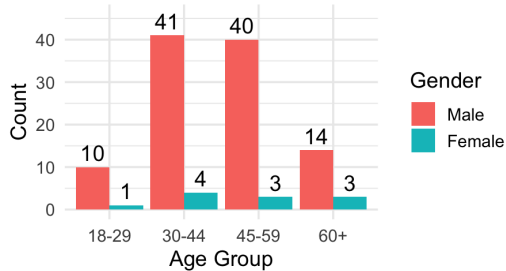
18-29 10.1% 30-44 39.5%  
 45-59 35.7% 60+ 14.7%

*The sample is predominantly male, with the highest representation in the 30–59 age range.*

## Demographic Cross-Tabulations

### Gender × Age Group

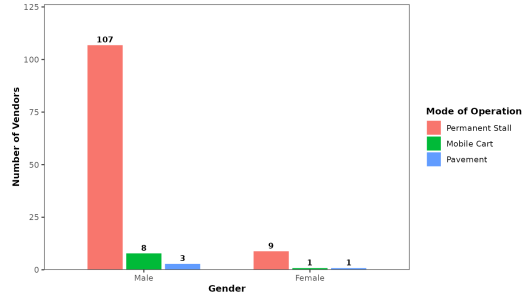
Gender Distribution Across Age Groups



Female participation remains limited across all age groups, with the highest representation in the 30–44 category.

### Gender × Mode of Operation

Gender .. Primary Mode of Operation ... Count  
n = 129 valid responses

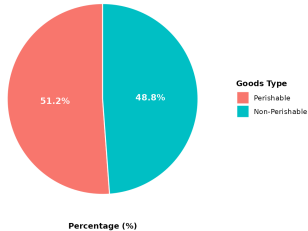


Permanent stalls dominate for both genders, while male vendors exhibit slightly greater operational diversity.

# Vendor Composition

## Goods Type

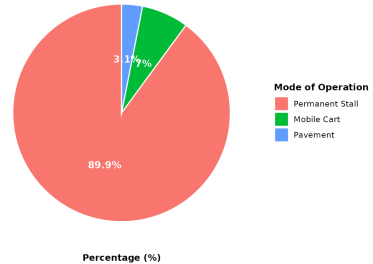
Type of Goods Sold by Street Vendors  
n = 129 respondents



Perishable **51.2%**  
Non-Perishable **48.8%**

## Operational Structure

Primary Mode of Operation of Street Vendors  
n = 129 respondents

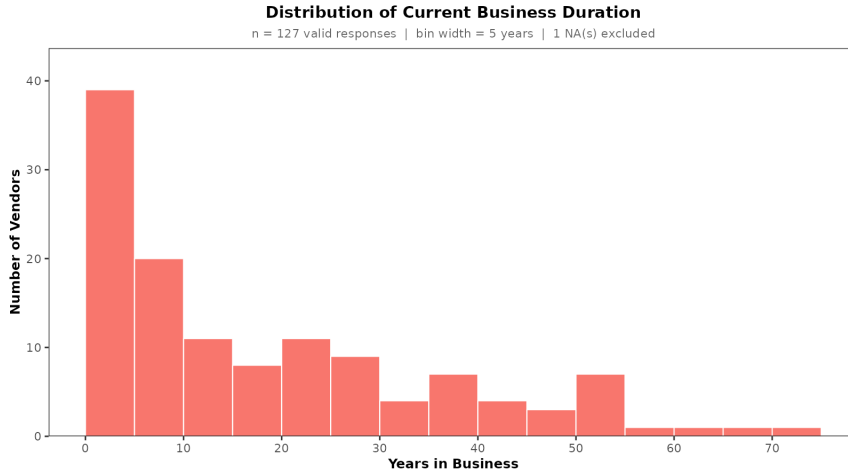


Permanent stalls dominate the surveyed corridor, with smaller representation from carts and pavement vendors.



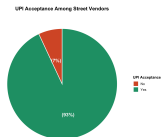
# Business Duration Distribution

## Raw Distribution of Business Duration ( $n = 127$ )



# UPI Acceptance and Adoption Drivers

## UPI Acceptance



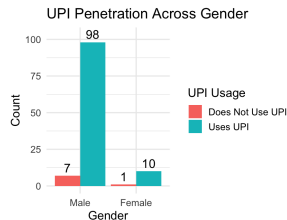
Accepted **93%**  
Not Accepted **7%**

## Adoption Drivers



Customer Demand **52.8%**  
Avoid Change **20%**  
Peer Influence **16%**

## UPI × Gender

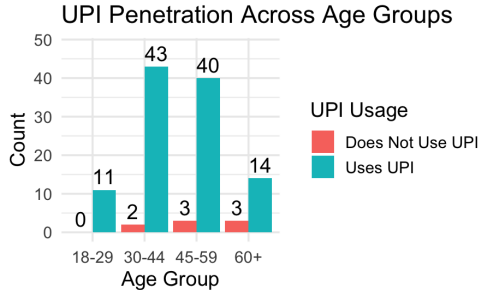


Uses UPI **93%**  
Does not use UPI **7%**

UPI adoption appears nearly universal, with customer demand emerging as the dominant motivating factor.

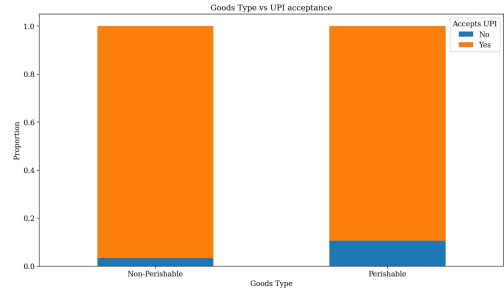
# UPI Penetration Across Demographic Groups

## Age × UPI Acceptance



UPI penetration remains high across all age groups, with complete adoption observed in the 18–29 category within the surveyed sample.

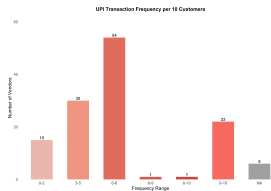
## Goods Type × UPI Acceptance



Non-perishable vendors exhibit near-universal UPI acceptance, while perishable sellers show slightly lower adoption rates.

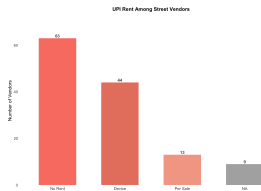
# Transaction Intensity and Operational Frictions

## UPI Transactions



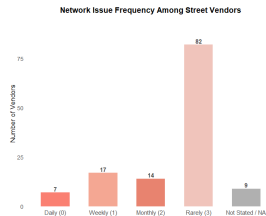
|      |    |
|------|----|
| 6-8  | 54 |
| 9-10 | 22 |
| 0-5  | 45 |

## UPI Fee Structure



|              |    |
|--------------|----|
| No Rent      | 63 |
| Device Rent  | 44 |
| Per-Sale Fee | 13 |

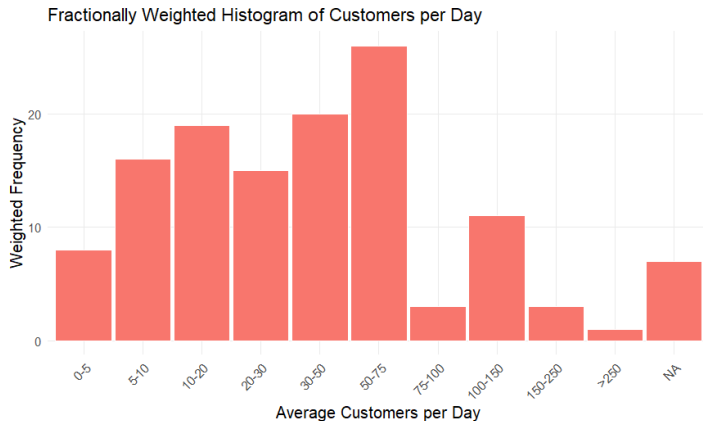
## Network Reliability



|         |    |
|---------|----|
| Rarely  | 82 |
| Weekly  | 17 |
| Monthly | 14 |
| Daily   | 7  |

Most vendors report moderate-to-high UPI usage, limited operational costs, and relatively infrequent network disruptions.

## Customer Flow: Last Hour and Daily Patterns (Contd.)

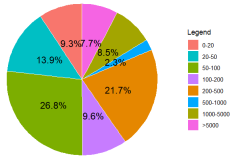


### Fractionally Weighted (Daily)

Peak at 50–75 customers/day. Broad cluster 10–75/day. Secondary surge at 100–150.

# Ticket Size Distribution

Pie Chart showing Distribution of Ticket Sizes



Log-Scaled Distribution of highest Earnings (Last 7 Days)

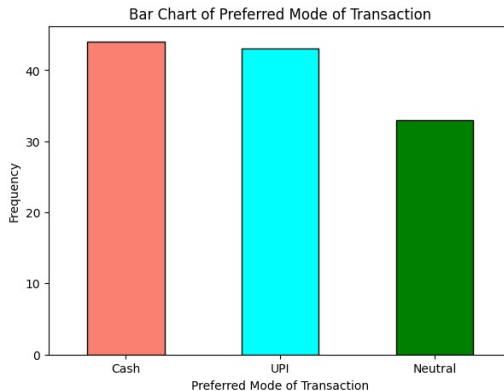


Log-Scaled Distribution of lowest Earnings (Last 7 Days)



*Ticket sizes: 50–100 (26.8%), 200–500 (21.7%). Majority of business driven by tickets between 50 and 500.*

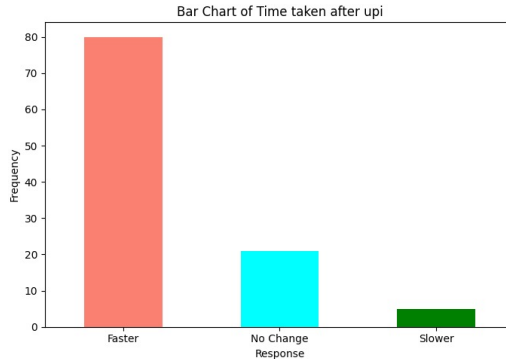
# Cash, UPI, and Behavioural Dynamics



## Preferred Mode

Cash: 44 — UPI: 43 — Neutral: 33

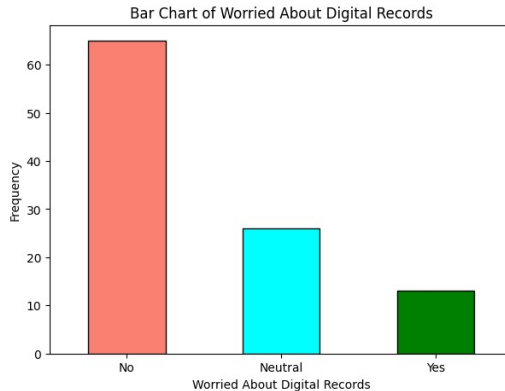
## Cash, UPI, and Behavioural Dynamics (Contd.)



**Transaction Speed**  
Faster: 80 (75%)  
No Change: 21 — Slower: 5



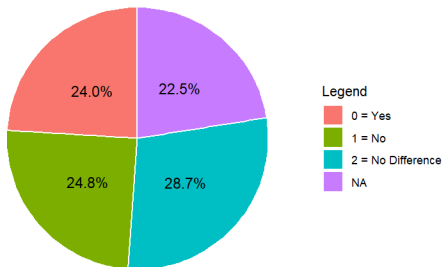
## Cash, UPI, and Behavioural Dynamics (Contd.)



**Digital Record Worry**  
No: 65 — Neutral: 26 — Yes: 13

## Cash, UPI, and Behavioural Dynamics (Contd.)

Pie Chart showing perception of people if UPI helps save more

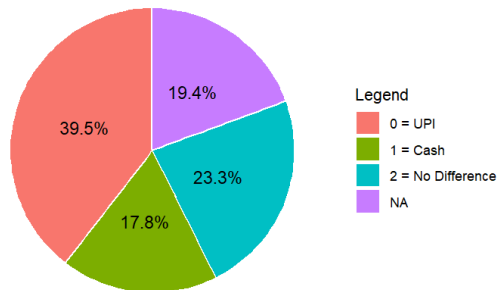


### Does UPI Help Save?

No Difference: 28.7% — Yes: 24.0% — No: 24.8% — NA: 22.5%

## Cash, UPI, and Behavioural Dynamics (Contd.)

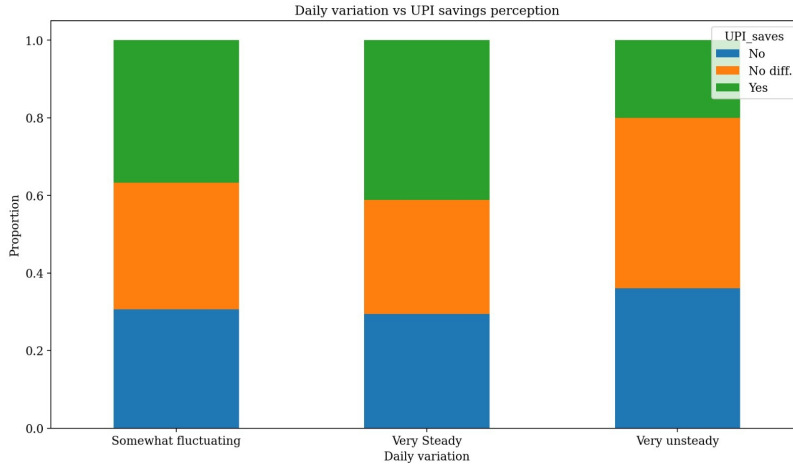
Pie Chart showing distribution of Payment Methods



### Payment Method Distribution

UPI: 39.5% — Cash: 17.8% — No Difference: 23.3% — NA: 19.4%

# Income Stability and UPI Savings Perception



## Income Stability and UPI Savings Perception (Contd.)

| Income Type          | Sentiment toward UPI Savings                                   |
|----------------------|----------------------------------------------------------------|
| Very Steady          | Most optimistic; largest “Yes” group                           |
| Somewhat Fluctuating | Middle ground; balanced between “Yes” and “No”                 |
| Very Unsteady        | Most skeptical; highest proportion of “No” and “No difference” |

*Vendors with steadier incomes are more likely to believe UPI helps them save.*

# Inferential Analysis

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## UPI Intensity Index: Motivation

- **Problem:** Short-run observations ( $\sim 1$  hour) are noisy; long-run recall is biased.
- **Goal:** Construct a statistically stable, vendor-level measure of UPI adoption intensity.
- **Solution:** Reliability-weighted Bayesian estimator combining:
  1. Coarse ordinal recall of long-run UPI usage (Question 12).
  2. Direct observational counts of UPI transactions in the last hour.

### Key Idea

Use an internal consistency check to down-weight unreliable recall responses.

## Observed Data Structure

- For vendor  $i$ :
  - $n_i$  = total number of customers observed in the last hour
  - $Y_i$  = number of customers who paid via UPI
- Response categories are mapped to numerical midpoints  $m_i$ :

| Response Category | Midpoint $m_i$ |
|-------------------|----------------|
| 0–2               | 0.10           |
| 3–5               | 0.40           |
| 6–8               | 0.70           |
| 9–10              | 0.95           |

- The midpoint  $m_i$  acts as a noisy prior estimate of the probability that a randomly arriving customer pays via UPI.



## Bayesian Model Specification

Let  $p_i$  denote the true UPI usage probability for vendor  $i$ . Then we assume that,

$$Y_i \mid p_i \sim \text{Binomial}(n_i, p_i)$$

$$p_i \sim \text{Beta}(s_i m_i, s_i(1 - m_i))$$

where  $m_i$  is the midpoint of the recalled usage bucket and  $s_i$  is the *prior reliability weight*.

| Consistency Level   | Interpretation       | $s_i$ |
|---------------------|----------------------|-------|
| Exact match         | High reliability     | 10    |
| Slight discrepancy  | Moderate reliability | 5     |
| Major contradiction | Low reliability      | 2     |

## Posterior Estimator & Properties

By Beta–Binomial conjugacy, the posterior mean is:

$$\hat{U}_i = \frac{s_i m_i + Y_i}{s_i + n_i} = \underbrace{\frac{s_i}{s_i + n_i}}_{\text{recall weight}} m_i + \underbrace{\frac{n_i}{s_i + n_i}}_{\text{data weight}} \frac{Y_i}{n_i}$$

### Advantages

- Regularises small-sample noise
- Avoids degenerate 0% or 100% estimates
- Discounts inconsistent recall

### Example

- Recall: 6–8 ( $m_i = 0.70$ )
- Slight discrepancy ( $s_i = 5$ )
- Observed:  $n_i = 4$ ,  $Y_i = 2$
- $\hat{U}_i = \frac{3.5+2}{9} \approx 0.611$

## Bayesian Estimation of Revenue Instability

**Problem:** Only recalled weekly revenue extrema are available (no daily panel).

Define the observed log-range:

$$R_i = \log\left(\frac{X_{i,\max}}{X_{i,\min}}\right), \quad R_i \geq 0$$

where  $(X_{i,\max})$  and  $(X_{i,\min})$  are the highest and lowest daily revenues reported for the preceding week.

Model structure (Gamma–Gamma):

$$\begin{aligned} R_i \mid \phi_i &\sim \text{Gamma}(k, \phi_i) \\ \phi_i &\sim \text{Gamma}(\alpha_{s_i}, \beta_{s_i}) \end{aligned}$$

where  $s_i \in \{0, 1, 2\}$  is the self-reported stability category.

## Bayesian Instability Index

Hyperparameters are initialised from empirical medians  $m_s$  within each stability category:

$$\mu_s = \frac{k}{m_s}, \quad \alpha_s = \lambda \mu_s, \quad \beta_s = \lambda$$

The **Bayesian Instability Index** is the reciprocal of posterior mean precision:

$$\hat{\theta}_i = \frac{\beta_{s_i} + R_i}{\alpha_{s_i} + k} = \frac{\lambda + R_i}{\lambda \mu_{s_i} + k}$$

- Larger  $\hat{\theta}_i \Rightarrow$  greater revenue instability.
- Shrinks extreme weekly recalls toward the category mean.

## UPI Adoption: Research Question

**Central Question:** Is there a statistically significant relationship between UPI intensity and vendor characteristics?

The Bayesian index  $\hat{U}_i$  is binarised:

$$\hat{U}_i \geq 0.5 \Rightarrow \text{High Intensity}, \quad \hat{U}_i < 0.5 \Rightarrow \text{Low Intensity}$$

Variables examined:

1. Age group
2. Primary mode of operation
3. Business trajectory
4. Type of goods sold
5. Business duration
6. Customer volume & ticket size

## Socio-Demographic Associations

| Variable            | Test             | Statistic       | p-value |
|---------------------|------------------|-----------------|---------|
| Age group           | Pearson $\chi^2$ | $\chi^2 = 5.03$ | 0.1694  |
| Mode of operation   | Fisher's Exact   | –               | 0.2366  |
| Business trajectory | Pearson $\chi^2$ | –               | 0.2895  |
| Type of goods       | Pearson $\chi^2$ | $\chi^2 = 2.21$ | 0.1367  |
| Business duration   | Mood's Median    | –               | 0.0989  |

### Finding

**None** of the socio-demographic or structural variables show statistically significant association with UPI intensity at the 5% level once the Bayesian estimator is used.

# Operational Characteristics

## Mann–Whitney $U$ Tests:

| Metric           | $U$ Statistic | p-value |
|------------------|---------------|---------|
| Customer Avg Day | 2335.0        | 0.0150  |
| Ticket Size      | 2701.0        | 0.0023  |

### Customer Volume

- **Negative** association with UPI intensity
- High-footfall vendors prefer cash for speed during peak hours

### Transaction Size

- **Positive** association with UPI intensity
- Large cash transactions carry logistical & security risks

## Logistic Regression: UPI Intensity

| Variable                | Coefficient    | p-value      | VIF         |
|-------------------------|----------------|--------------|-------------|
| Gender                  | 0.6198         | 0.425        | 1.04        |
| Age Group               | -0.3556        | 0.182        | 1.14        |
| Type of Goods           | 0.4929         | 0.273        | 1.20        |
| Business Duration       | -0.0239        | 0.059        | 1.19        |
| Mode of Operation       | -0.7483        | 0.203        | 1.09        |
| <b>Customer Avg Day</b> | <b>-0.0142</b> | <b>0.039</b> | <b>1.30</b> |
| Daily Variability       | -0.0016        | 0.996        | 1.10        |
| Network Issue           | -0.3270        | 0.210        | 1.15        |

Model LLR  $p$ -value = 0.0092. All VIFs < 5.

**Takeaway:** High UPI intensity is driven by *transaction profile* (ticket size), not business scale (footfall).



## Summary: UPI Adoption

- Apparent generational or structural patterns largely vanish under the reliability-weighted Bayesian index.
- UPI adoption is **heterogeneous** and **not strongly tied** to observable demographics.
- The strongest operational predictor is **transaction size** (+), while **customer volume** is unexpectedly negative.
- Simple raw-intensity measures likely confound measurement noise with true structural effects.

## Motivation & Monetisation

### Objective:

1. Quantify sales lost due to cash change-shortage.
2. Examine whether higher UPI intensity mitigates this loss.

Estimated daily revenue loss:

$$RL_i = CLC_i \times TS_i$$

where  $CLC_i$  = customers lost due to change shortage and  $TS_i$  = average transaction size.

| Statistic              | Value (Rs.) |
|------------------------|-------------|
| Median daily loss      | 240.00      |
| Mean daily loss        | 759.92      |
| Annualised median loss | 87,600      |

## Regression Framework

Binary outcome (collapsed from sparse categories):

$$Y_i = \begin{cases} 0, & \text{Low loss (0–2 customers)} \\ 1, & \text{Elevated loss } (\geq 3 \text{ customers}) \end{cases}$$

Logistic regression:

$$\log\left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)}\right) = \beta_0 + \beta_1 \hat{U}_i + \mathbf{X}_i^\top \boldsymbol{\beta}$$

Both **unweighted** and **inverse-frequency weighted** models estimated. Weighted model emphasises the minority elevated-loss cases.

## Regression Results: Mitigation Effect

| Predictor                | Coef.   | SE     | p-value | OR     |
|--------------------------|---------|--------|---------|--------|
| <i>Unweighted Model</i>  |         |        |         |        |
| Bayesian UPI $\hat{U}_i$ | -2.3718 | 0.9883 | 0.0164  | 0.0933 |
| <i>Weighted Model</i>    |         |        |         |        |
| Bayesian UPI $\hat{U}_i$ | -2.2964 | 0.3458 | <0.001  | 0.1006 |
| Ticket size              | -0.0040 | 0.0017 | 0.0178  | 0.9960 |
| Business duration        | 0.0400  | 0.0149 | 0.0071  | 1.0408 |
| Daily variability        | 0.6868  | 0.3445 | 0.0462  | 1.9873 |

### Practical Interpretation

A 10-percentage-point increase in  $\hat{U}_i$  yields an odds ratio of  $\approx 0.79$ .

⇒ Lower digital engagement carries a **26.7% “friction penalty”** in customer attrition.

## Conclusion: Change Shortage

- Cash change-shortage imposes a **measurable, recurring cost**: median  $\sim$ Rs. 240/day.
- Higher **UPI intensity significantly reduces** the probability of elevated customer loss.
- The mitigating effect appears only when UPI is **integrated into everyday selling**, not merely available.
- UPI reduces dependence on exact cash denominations, preventing transaction failures.

## Revenue Volatility: Question & Index

**Question:** Does higher UPI intensity reduce day-to-day revenue instability?

**Index used:** Bayesian shrinkage instability  $\hat{\theta}_i$ .

Descriptive validation ( $n = 92$ ):

| Stability Category   | $n$ | Mean $\hat{\theta}_i$ | Median $\hat{\theta}_i$ |
|----------------------|-----|-----------------------|-------------------------|
| Very steady          | 21  | 0.174                 | 0.167                   |
| Somewhat fluctuating | 55  | 0.238                 | 0.241                   |
| Very unsteady        | 16  | 0.297                 | 0.288                   |

Kruskal–Wallis:  $H = 54.36$ ,  $p < 0.0001$  Internal validity confirmed.

## Exploratory Patterns

### Aggregate Relationship

- Spearman  $\rho_S = 0.135$ ,  $p = 0.200$
- Slightly **positive**, not negative
- Mann–Whitney:  $p = 0.909$

### Segmented by Ticket Size

- **Low ticket** (<Rs. 100): weak negative slope
- **Medium/High ticket**: no stabilising pattern

## Exploratory Patterns: Economic Intuition

### Economic Intuition

Low-ticket vendors face exact-change frictions → UPI helps.

High-ticket vendors face lumpy demand → instability is structural, not payment-related.



## Regression: Controlling for Goods & Location

| Variable                    | Coef.         | SE            | p-value      |
|-----------------------------|---------------|---------------|--------------|
| Constant                    | 0.2290        | 0.0159        | <0.001       |
| UPI intensity $U_i$         | 0.0074        | 0.0207        | 0.721        |
| <b>Non-perishable goods</b> | <b>0.0283</b> | <b>0.0119</b> | <b>0.017</b> |
| Location: Baranagar         | -0.0422       | 0.0232        | 0.069        |
| Location: GLT Road          | -0.0016       | 0.0162        | 0.924        |

### Null Finding

The UPI coefficient is close to zero and insignificant. **UPI intensity does not reduce overall revenue volatility** at the full-sample level.

**Sensitivity:** Across  $k \in \{3, 5, 10\}$  and  $\lambda \in \{2, 5, 10\}$ , UPI  $p$ -values always  $> 0.30$ .

## Conclusion: Revenue Volatility

- **No evidence** that UPI intensity reduces daily revenue volatility.
- Revenue instability is driven by **business structure**:
  - Non-perishable goods → higher volatility (episodic, high-ticket demand)
  - Ticket size → lumpy sales dominate log-range
- UPI solves *transaction friction*, not *demand uncertainty*.

## Perception Dimensions

Analysis of vendor attitudes across:

1. Transaction speed vs. cash
2. Savings behaviour & financial tracking
3. Digital trust / worry about records
4. Network issues & payment-mode preference
5. Customer loss due to UPI problems

Statistical approach: Fisher's Exact Test or Pearson  $\chi^2$  (5% threshold).

## Key Perception Results

| Question               | Segmentation        | p-value |
|------------------------|---------------------|---------|
| Transaction speed      | Age group           | 0.668   |
| Saves more (bank)      | Age group           | 0.7167  |
| Saves more (bank)      | Goods type          | 0.8188  |
| Saves more (bank)      | Mode of operation   | 0.7803  |
| Security concern       | Age group           | 0.928   |
| Network vs. preference | –                   | 0.669   |
| UPI-related loss       | Mode of operation   | 0.3398  |
| UPI-related loss       | Business trajectory | 0.1219  |

### Uniform Acceptance

**No significant association** with age, goods type, or mode. UPI is perceived as **faster than cash** across *all* groups.

## Perception: Business Experience

- All vendors who **often** lost customers due to UPI issues reported business trajectory as **“Same”** (not grown).
- Kruskal–Wallis on **high earnings** by UPI-loss category:  $p = 0.0266$ .
- Low earnings: not significant ( $p = 0.3375$ ).

### Interpretation

Repeated digital friction may limit **upside potential** (best-day performance) even if it does not immediately cause decline.

# Primary Motivating Factors

## Demand-led adoption:

1. **Customer demand** (70 vendors)
2. Avoid change problem (28)
3. Peer / social influence (24)
4. External events (7)

$\chi^2$  test (shop age vs. motive):  $p = 0.468$   
 $\Rightarrow$  Customer demand cuts across all ages.

## Fee Arrangements

- No rent: 58 (53.7%)
- Device rent: 37 (34.3%)
- Per-sale rent: 13 (12.0%)

## Multinomial Logistic Regression: Fee Arrangement

Outcome:  $\text{upi\_rent}_i \in \{\text{No rent, Device, Per-sale}\}$ .

Reference: No rent. Predictors: business duration (ordered) & ticket size (ordered).

| Outcome       | Predictor        | Coef.   | p-value | RRR    |
|---------------|------------------|---------|---------|--------|
| Device rent   | Bus. duration .Q | 0.8788  | 0.0333  | 2.4080 |
| Device rent   | Ticket size .L   | -1.2690 | 0.0290  | 0.2811 |
| Per-sale rent | Bus. duration .Q | 1.6154  | 0.0194  | 5.0300 |

- **Business duration:** Non-linear. Middle-duration group has *lowest* fee prevalence; oldest group has *highest*.
- **Ticket size:** Negative linear effect on device rent. Larger-ticket vendors less likely to pay device rent (RRR=0.281).

## Conclusion: Motivations & Fees

- UPI adoption is **reactive**, not purely voluntary:
  - Driven by **customer pressure**, convenience, and peer effects.
- Fee structure is linked to **business maturity** and **transaction scale**:
  - Very mature businesses → more fee-based arrangements.
  - Larger-ticket vendors → less device rent.
- No significant link between fee arrangement and business trajectory ( $p = 0.7208$ ).



## Part IV: Synthesis

### What UPI Does Do

- Reduces cash change-shortage friction
- Improves transaction speed (perceived)
- Becomes default infrastructure for micro-payments

### What UPI Does Not Do

- Eliminate daily revenue volatility
- Correlate strongly with demographics
- Stabilise lumpy, high-ticket demand

### Core Message

UPI is an indispensable **layer of transactional infrastructure**, not a transformation of informal retail fundamentals.

## Conclusions

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## Concluding Synthesis

### What this study examined

- Role of UPI in day-to-day functioning of informal-sector vendors along the B.T. Road corridor
- Primary survey data from **129** street vendors
- Not just *extent* of adoption, but **economic and behavioural consequences**

### Core question

Does UPI adoption merely reflect a change in payment mode, or does it produce **measurable operational benefits** for cash-constrained vendors?

## Concluding Synthesis: Key Descriptive Facts

### Key descriptive facts

- Acceptance is **widespread** (93%)
- Relevant variation now lies in **intensity of use**, not basic adoption
- Vendor population: male-dominated, 30–59 age range, permanent stalls
- Balanced perishable / non-perishable split
- Digital payments have moved from innovation to **standard commercial practice**

## Key Empirical Finding 1: UPI Intensity Patterns

### Significant predictors of UPI intensity

- **Age group** ( $p = 0.024$ )
- **Business duration** ( $p = 0.009$ )

Younger vendors and newer businesses tend to show **higher** levels of digital-payment use.

### Non-significant predictors

- Customer volume ( $p = 0.268$ )
- Type of goods sold ( $p = 0.568$ )
- Mode of operation ( $p = 0.215$ )
- Business trajectory ( $p = 0.290$ )

## Key Empirical Finding 1: Interpretation

### Interpretation

Adoption is shaped more by **adaptability**, **business age**, and **transactional habits** than by business scale or form.

### Takeaway

UPI intensity reflects the vendor's *willingness and ability* to integrate digital payments into everyday practice — not simply the size or structure of the business.

## Key Empirical Finding 2: Cash Change-Shortage Problem

### The problem

- Vendors often lose sales when they cannot provide exact change
- Median daily loss: Rs. **240**
- Annualised median loss: Rs. **87,600**
- Highly right-skewed: some vendors lose much more

### The result

- Higher UPI intensity is **strongly associated** with lower customer-loss severity
- Weighted logistic regression: OR  $\approx$  **0.10**
- $p < 0.001$

## Key Empirical Finding 2: Interpretation

### Why this matters

UPI is valuable not merely as an *alternative payment channel*, but as a **practical mechanism for reducing transaction frictions**.

### Mechanism

In high-frequency retail with small transactions, digital payment capability helps prevent otherwise avoidable sales failures.



## Key Empirical Finding 3: Revenue Volatility

### What the analysis shows

- **No strong evidence** that UPI intensity reduces overall daily revenue volatility
- Spearman  $\rho_S = 0.135$ ,  $p = 0.200$
- OLS coefficient on UPI intensity:  $0.0074$ ,  $p = 0.721$
- Sensitivity analysis: robust across all hyperparameter choices

### What does matter

- **Goods type** (non-perishable = higher instability)
- **Ticket-size patterns**
- Business structure, not payment mode

## Key Empirical Finding 3: Interpretation

### Crucial distinction

UPI can reduce **operational friction** at the point of sale, but it does not eliminate broader instability arising from demand variation, product type, or market structure.

### Conclusion

The effect of UPI is **narrower and more targeted** than a general income-stabilizing tool.

## Key Empirical Finding 4: Perceptions and Adoption Drivers

### Perceptions are normalised

- Most view UPI as **faster** than cash (75%)
- Not confined to any single age group or business type
- Attitudes toward savings, tracking, and trust are **mixed but uniform**
- Digital trust is high **across all demographics**

## Key Empirical Finding 4: Perceptions and Adoption Drivers (Contd.)

### No significant segmentation

- Transaction speed  $\times$  age:  $p = 0.668$
- Security concern  $\times$  age:  $p = 0.928$
- Savings perception  $\times$  age/goods/mode: all  $p > 0.7$

## Key Empirical Finding 4: Perceptions and Adoption Drivers (Contd.)

### Adoption is market-led

| Driver               | Vendors |
|----------------------|---------|
| Customer demand      | 70      |
| Avoid change problem | 28      |
| Peer influence       | 22      |

Vendors use UPI because **customers expect it** and because it improves transaction convenience.

## Broader Interpretation

### What UPI does well

- Reduces **everyday transactional frictions**
- Completing payments without exact cash is economically meaningful
- Small frictions, repeated many times daily, have **cumulative effects** on sales and retention
- Especially valuable for frequent, low-value transactions

### What UPI does not do

- Does **not** transform business fundamentals
- Does **not** solve broader uncertainties of informal-sector commerce
- Vendors may become digitally integrated while still facing unstable earnings and irregular demand
- Digital payment  $\neq$  complete economic stabilization

### Bottom line

UPI improves the **mechanics of exchange**; the deeper structure of informal economic life remains shaped by demand, business type, and market conditions.

## Methodological Contribution

### Extracting evidence from small, noisy field data

This study demonstrates that useful statistical evidence can be drawn from informal-sector surveys when analysis is designed carefully.

#### Non-parametric tests

- Fisher's Exact Test
- Mood's Median Test
- Mann–Whitney  $U$
- Kruskal–Wallis

For skewed distributions and small cell counts

#### Regression frameworks

- Binary logistic regression
- Inverse-frequency weighting
- OLS with robust SEs
- Multinomial logit

For mitigation and structural analysis

#### Bayesian methods

- Bayesian UPI intensity estimator
- Shrinkage volatility index
- Category-specific Gamma priors

For recalled data and noisy measures

*Especially valuable where direct measurement is difficult and perfect data are rarely available.*

# Limitations

## 1. Cross-sectional design

- Results should not be interpreted as *causal* estimates
- No temporal ordering of adoption and outcomes

## 2. Self-reported and recall-based measures

- Earnings, customer counts, and volatility indices rely on respondent recall
- Potential measurement error and perception bias

## 3. Geographic restriction

- Sample confined to the B.T. Road corridor
- Findings may not generalise to other urban or rural markets

## 4. Approximation and grouping

- Interval-based responses required midpoint imputation
- Some quantitative constructions are approximate

*These limitations suggest caution in interpretation and provide clear directions for future work.*



## Scope for Future Research

### Design improvements

- **Longitudinal design** to track UPI impact over time
- Combine vendor-side data with **customer-side evidence**
- Distinguish supply-side adoption from demand-side preference

### Additional topics

- Seasonal effects on revenue and UPI usage
- Network reliability and transaction failure rates
- Digital payments and access to credit / savings

## Scope for Future Research: Comparative and Causal Work

### Comparative and causal work

- **Comparative studies** across different corridors or cities
- Are B.T. Road patterns specific or part of a broader trend?
- **Randomised encouragement designs** for causal inference

### Key question

Does increasing UPI intensity lead to measurable changes in sales stability, income growth, or customer retention **over time**?

## Final Remarks

### UPI's role in informal retail

- An **important and practical** feature of commerce
- Strongest effect: reducing cash-related transaction friction
- Especially where small payments and exact change matter

## Final Remarks (Contd.)

### What UPI is not

- Not a **universal solution** to revenue instability
- Not a **complete transformation** of informal retail
- Does not alter the deeper structure of demand, business type, or market conditions

## Final Remarks (Contd.)

### Final Word

UPI should be understood not as a complete transformation of informal retail, but as a **meaningful and increasingly indispensable layer of transactional infrastructure.**

# Thank You

Questions & Discussion

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## Appendix: Categorical Encoding

| Variable          | Categories                  | Code          |
|-------------------|-----------------------------|---------------|
| Gender            | Male / Female               | 1 / 2         |
| Age Group         | 18–29 / 30–44 / 45–59 / 60+ | 1 / 2 / 3 / 4 |
| Goods Type        | Perishable / Non-perishable | 0 / 1         |
| Mode of Operation | Stall / Cart / Pavement     | 0 / 1 / 2     |
| UPI Acceptance    | No / Yes                    | 0 / 1         |
| UPI Rent          | No / Device / Per-sale      | 0 / 1 / 2     |

## Appendix: Selected References

1. NPCI (2016). *Unified Payments Interface*.
2. RBI (2023). *Payment Systems in India*.
3. BIS (2021). *Digital Payments and Financial Inclusion*.